Visvesvaraya National Institute of Technology, Nagpur
Department of Mechanical Engineering

The v3 Controller/Estimator Campaign on the Mecanum Digital-Twin Plant

A fair three-way control-law comparison — **PID-FB / PID-CT / ASMC v2** — on a voltage-input slip–spin–LuGre plant, through a frozen ESKF, with feasibility-screened trajectories and paired statistics.

Vishvesh Koranne · Research Scholar
Guide: Prof. Shital S. Chiddarwar

KUKA youBot · slip–spin digital twin
VNIT Nagpur · August 2026

Outline

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One question, three control laws, one scalar objective

Which control law should the platform fly — and how much survives real state estimation?

Controller	Structure	What it isolates
PID-FB	Cascaded IMC-reparameterized PID — 3 tuned λ_{inner} per axis, everything else derived from the plant model (pole cancellation, I_{max} from measured p95 wrench)	industrial baseline
PID-CT	Same cascade **+ computed-torque feedforward**, structurally matched to ASMC's equivalent control	FB→CT: value of the **model**
ASMC v2	Adaptive-gain supertwisting SMC — friction-circle gain ceiling K_{max} recomputed per tick, smooth growth gate	CT→ASMC: value of **adaptation**

FAIRNESS DISCIPLINE

Same optimizer, same objective, same trajectory tiers, same frozen estimator. Anything that makes one controller's objective differ from another's is treated as a **bug, not a detail**. MPC deferred this iteration (diverges on 4 velocity-demanding trajectories; use_ltv anomaly unresolved).

The plant is voltage-input: back-EMF is the hard wall

30-state slip–spin–LuGre plant (per-wheel bristle states, passive roller spin) driven through a DC-motor map evaluated every integration step:

$$\tau = G\eta\,(K_t i - \tau_f), \qquad i = \frac{V - K_b\,G\,\omega}{R_a}, \qquad |V| \le 24\ \text{V}$$

- Torque falls linearly with wheel speed; **stall 9.12 N·m**, slope 0.337 N·m·s/rad
- **No-load wall** $\omega_{nl} = V_{max}/(K_b G) = \mathbf{27.73}$ rad/s — torque authority vanishes identically
- Caps: $V_x \le \mathbf{1.02}$ m/s (73.5% nl), $\dot{\psi} \le \mathbf{2.47}$ rad/s *(motor-limited)*; $V_y \le \mathbf{0.63}$ m/s *(friction-limited)*

:REALISTIC SENSOR SUITE

- **IMU** 1000 Hz — σ_{acc} 0.05 m/s², σ_{gyro} 3 mrad/s, biases, gyro RW
- **Encoders** 1000 Hz — 4000 cpr × gear, $\sigma_{\omega} \approx$ 0.19 rad/s, 2% scale factor
- **Optical flow** 100 Hz — σ 5 mm/s + 2% of speed, 2% dropout
- **Pose fix** 100 Hz — σ_{pos} 0.02 m, σ_{ψ} 0.6°, per-run biases, 0.5% outliers

NEITHER SIDE SEES TRUTH

The estimator never sees the true state; in the deployable configuration the **controllers** don't either — feedback is the ESKF output.

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Three feedback regimes — not two

Conflating them inverts conclusions; every number in the campaign is tagged by regime

CLEAN ORACLE

Perfect state feedback.

An **upper bound** on what any controller can do — the reference for what estimation costs.

NOISY ORACLE

True state **+ raw unfiltered sensor noise** written straight into the controller's state estimate.

No estimator exists in this path — a pessimistic no-filtering bound, not a deployment model.

ESKF (DEPLOYABLE)

:realistic sensors through the frozen 13-state error-state Kalman filter (yaw-acceleration state).

Beats the noisy oracle **by construction** — a tuned filter beats raw noise. The cost of estimation is ESKF vs **clean**, never ESKF vs noisy-oracle.

Feasibility first: two walls, evaluated per reference

train12 → train14_v3 / test_v3 — a comparison is only meaningful where the hardware can go

WALL 1 — MOTOR / BACK-EMF (HARD)

Max per-wheel speed $(V_x \pm V_y \pm (l{+}h)\dot{\psi})/R$ against $\omega_{nl} = 27.73$ rad/s. Past ≈85% of no-load no torque exists to arrest error growth, for **any gain law**. v3 targets ≤70% for margin.

WALL 2 — FRICTION CIRCLE (SOFT)

Per-wheel contact utilisation. Brief excursions are momentary slip the LuGre plant absorbs — measured harmless (both >1.0-util octagons are among the *best-tracked*). Informs, does not veto.

WHY THE TIER HAD TO BE REBUILT

- **spiral_orbit_stress**: 85.4% of no-load, carried 72–81% of the training tracking term — the tuner was optimising against a trajectory no controller recovers
- Old screen ($V_y \leq 0.6$) saw **neither** infeasible case — both had $V_y = 0.000$
- Every reference now evaluated over its **own full** T_{total} (9.5–85.3 s) — a 12 s audit window once picked a combo that is 98.5% of no-load over its true duration

Worst-case wheel speed: train12 85.4% → train14_v3 67.3% of no-load.

One scalar: tracking + energy + priced chatter

$$\text{score} = \text{tracking}_{v3} + 0.05\,\frac{ce}{V_{MAX}} + 0.36\,\frac{chatter}{CHATTER_{REF}}, \qquad CHATTER_{REF} = 0.8\ \mathrm{V/ms}$$

- **tracking$_{v3}$** — 6 tolerance-normalised terms (position & heading × *terminal* / *peak* / *typical*), divided by a rotation-invariant per-trajectory extent scaler k_{traj}. Term = 1.0 means "exactly at target".
- v2 scored **4 samples out of 12k–60k** (terminal + argmax) — blind to mid-path oscillation. The time-normalised integral ("typical") is the fix; tolerances **calibrated**, not chosen.
- **chatter** priced against the mixer's own slew clamp — $\lambda = 0.36$ is **balance-preserving** (reproduces v2's accepted 29.9% tracking share), not re-derived; the code *refuses* $\lambda > 1.44$ under v3.

WHY CHATTER NEEDED PRICING

Chatter had been normalised by V_{MAX} — a voltage dividing a slew rate, **~30–100× too weak**. Three v2 box-widening rounds kept railing on a lever the objective could not price. A term that cannot move the score is worse than no term.

ESTIMATOR'S OWN OBJECTIVE

Replay vs ground truth, same tolerance-normalised shape; velocity error **during slip priced twice** — model mismatch on this platform *is* slip. Converges to ~71% a velocity-estimation objective — what inner loops consume.

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Estimator first, then frozen — then controllers

1 · Tune ESKF (replay, 5 seeds)
7-dim log space · :realistic sensors · seed spread = noise artefact (0.23% on common realisation)

→

2 · Freeze seed 4
never re-tuned per tier — a control-law difference can never be an observer difference in disguise

3 · Clean tuning
5 seeds × 3 controllers · oracle-clean · train14_v3 · plateau-detected convergence (v1 was caught mid-descent)

→

4 · Noisy tuning
4 seeds × 3 controllers · warm-start from own clean optimum · noisy-oracle feedback · all three or none

5 · Evaluate 6 configs × 3 regimes
fresh sensor seeds 101–105 · clean / noisy-oracle / frozen-ESKF · 420 closed-loop runs, zero non-converged

→

+ test_v3 leg ✓
held-out generalisation, 7 unseen trajectories (3 unseen profiles): **PID-CT wins 35/35 paired vs each rival**

One OS process per seed (global mutable state) · BLAS pinned to 1 thread (determinism) · stop_reason recorded · runs_*/ never hand-edited

Oracle feedback: PID-CT wins clean and noisy

Lower score = better · eval on fresh sensor seeds 101–105 · paired statistics throughout

Config	clean score	noisy-oracle score
PID-CT clean-tuned (s5)	**0.14340**	0.50372
PID-CT noisy-tuned (s4)	0.14422	**0.49799**
PID-FB clean-tuned (s3)	0.14953	0.53135
PID-FB noisy-tuned (s4)	0.15492	0.53199
ASMC clean-tuned (s2)	0.15318	0.53168
ASMC noisy-tuned (s4)	0.15697	0.51838

Clean ordering **PID-CT < PID-FB < ASMC**, no family overlap, seed spread 0.15–0.5%.
Noisy-oracle ordering **PID-CT < ASMC < PID-FB**, 4/4 paired.

NOISY TUNING HELPS — ON ITS OWN REALISATION

Paired per realisation: ASMC **5/5 better**, PID-CT **5/5 better**, PID-FB mixed (structural — no feedforward, integrator at its anti-windup bound on 20.5% of ticks, chatter already at 0.758 of the 0.8 cap: nowhere left to go).

BUT SEED = NOISE DRAW

Identical seed ordering for all three controllers (s4<s2<s3<s1): the ~16% tuning seed spread is **realisation variance**, not quality. Rankings are always paired, never marginal means.

Through the frozen ESKF: the ranking holds

420 runs (6 configs × 14 trajectories × 5 noise seeds), zero non-converged

Config	ESKF score	track	chatter
PID-CT ct(s5)	**0.34127**	0.02393	0.532
PID-CT nt(s4)	0.34341	0.02624	0.532
PID-FB ct(s3)	0.36054	0.03315	0.554
PID-FB nt(s4)	0.36298	0.03149	0.563
ASMC nt(s4)	0.36320	0.04128	0.542
ASMC ct(s2)	0.36423	0.03658	0.554

Paired per (trajectory, seed), n=70: **PID-CT beats ASMC 69/70, PID-FB 70/70** — best in all three regimes. ASMC vs PID-FB: 33/70 — **tied** (the noisy-oracle ASMC edge does not survive real estimation). On the held-out test_v3 tier: **PID-CT wins 35/35 against each rival** — the campaign's most robust result.

COST OF REAL ESTIMATION: +0.20, UNIFORM

ESKF vs clean oracle: +0.198 to +0.211 across all six configs; the filter recovers **42.9–45.1% of the clean→unfiltered gap**. The estimator's contribution is **controller-independent** — what a frozen observer is for.

NOISY TUNING DOES NOT TRANSFER

Its oracle-regime gains vanish through the ESKF, and it **damaged ASMC's heading ~2× everywhere**. Deploy clean-tuned configs.

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Three stages: only the tracking stage discriminates

Channel	Injected (sensors)	ESKF delivered	Removed
velocity	~13 mm/s	2.71 mm/s	~5×
position	~22 mm	3.25 mm	~7×
yaw rate	~4 mrad/s	2.33 mrad/s	~2× (weakest)
heading	~11 mrad	1.26 mrad	~9× (best)

- Stage (1) injection — **identical across configs by construction**
- Stage (2) ESKF survival — 74.2–78.5%, barely controller-dependent (frozen filter)
- Stage (3) tracking — **essentially all inter-controller spread lives here**

NOISE FLOOR LIMITS THE COMPARISON

On **7 of 14** trajectories, tracking error (~2–3 mm) sits at or below the estimator's own pose accuracy (~3.2 mm) — further controller improvement there is **unmeasurable** with this sensor suite. The comparison is decided by the few trajectories that rise clear of the floor.

HEADING'S "BAD MEAN" WAS AN ARTEFACT

Three references start at $\psi(0) \neq 0$; the ESKF initialises at 0 and converges in **11.0 ms** (same on all three, two different initial errors). Steady-state heading is the estimator's *best* channel.

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PID-CT's heading weakness: two trajectories, three measurements

Trajectory	PID-CT heading	PID-FB heading	CT yaw I-clamp
9 of 14 (benign)	**1.04–1.31 mrad (best)**	1.13–1.96	0.0% of ticks
coupled_vomega_stress	29.8–38.2 mrad	1.28	19.9%
spiral_orbit_stress	37.6–46.8 mrad	2.35	17.0%

Three independent measurements land on the same two sustained-yaw trajectories: HEAD-sum concentration, integral-clamp saturation fraction, and closed-loop heading tracking in physical units.

MECHANISM (CANDIDATE #1, CONFIRMED BINDING)

CT's I_{max} is **derived, not tuned** — $K_i I_{max}$ = measured wrench budget, sized at p95 of integral demand. On sustained yaw the clamp fires; whether binding *harms* (vs does its job) needs a budget-sensitivity run.

IT COMPOUNDS WITH THE ESTIMATOR

Yaw rate is the ESKF's weakest channel (~2× removal, gyro-only) **and** CT's weakest tracking channel — one fix (rate re-weighting or I_{max} re-derivation) attacks both.

Five rules that keep these numbers honest

Each rule exists because its violation produced a wrong conclusion at some point in the project

- **Never mix tuning-stage scores with eval scores** — different noise realisations, overlapping ranges.
- **Pair, don't average, across realisations** — realisation variance (~6–7% eval, ~16% tuning) swamps controller differences (~2–4%). Every ordering claim is a paired 5/5 or 4/4.
- **Best-seed selection under noise is confounded** — seed 4 is simultaneously the best seed and the easiest realisation for all three controllers (applied identically, so cross-controller comparison holds).
- **Absolute values, never % degradation** — a better clean tracker reads a *worse* % for the same absolute result.
- **Three feedback regimes, not interchangeable** — ESKF-minus-noisy-oracle is *not* "the cost of state estimation"; that cost is ESKF vs clean oracle. Getting this wrong inverts the sign of the conclusion.

Generalised to held-out trajectories — what remains open

- **test_v3 generalisation ✓** — 7 genuinely held-out trajectories (3 unseen profiles; the 8th is a deliberate anchor, bit-identical across tiers). **PID-CT: 35/35 paired vs each rival, first on every held-out trajectory.** ASMC vs PID-FB does *not* generalise — their training-tier gap was one trajectory; the defensible claim is **ASMC ≡ PID-FB, PID-CT beats both**.
- **Estimator is tier-invariant** — every error-budget channel within ~3 pp of its training value; all controllers track 1.6–4.2 mm / 1.0–2.5 mrad on held-out data.

- **CT yaw I_{max} budget sensitivity** — the clamp is measured to *bind*; that it *harms* is unmeasured.
- **ESKF yaw-rate re-weighting** — weakest channel (~2× removal), gyro-only; interacts with the CT heading finding.
- **$\hat{\psi}(0)$ from the first pose fix** — removes the 11 ms startup transient on non-zero-initial-heading references.
- **MPC** — deferred; the use_ltv anomaly (false better than true) remains unresolved from v2.
- **Downstream** — PINN $\hat{\gamma}$-observer tuning & force reconstruction consume this stack with noisy estimator-fed inputs (train-like-you-test), oracle retained as unit test and error budget.

HEADLINE

PID-CT (cascaded feedforward PID) is the deployment candidate — best in clean oracle, noisy oracle, through the frozen ESKF (69/70, 70/70), **and on held-out trajectories (35/35 vs each rival)** — with one mapped weakness: heading under sustained yaw.